5 Lecture 5 (January 22, 2025): Monoidal categories and braidings

Scribe: Jackson Morris

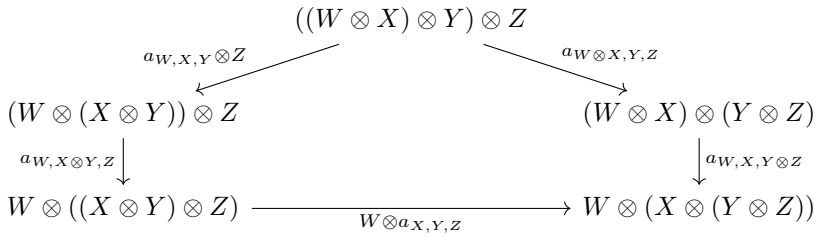
We now introduce monoidal categories and braidings. For further references, see [EGNO15] sections 2, 5, and 8. Last time we defined what an abelian k-linear category was, with the prototypical example being the category G-mod where G is an affine group scheme over k. Today, we abstract those definitions and build on them.

5.1 Definition. A *monoidal structure* on a category $\mathcal{C}$ consists of $(\otimes, a, \mathbb{1}, \iota)$, where:

- $- \otimes - : \mathcal{C} \times \mathcal{C} \to \mathcal{C}$ is a bifunctor;
- $a_{X,Y,Z} : (X \otimes Y) \otimes Z \to X \otimes (Y \otimes Z)$ is a natural isomorphism,

such that the following holds:

1. The following diagram commutes:



   which we call the *pentagon axiom*;

2. There is a pair $(\mathbb{1}, \iota)$ where $\mathbb{1} \in \mathrm{ob}\mathcal{C}$ and

$$\iota : \mathbb{1} \otimes \mathbb{1} \to \mathbb{1}$$

   is an isomorphism such that the functors

$$L_{\mathbb{1}} : X \mapsto \mathbb{1} \otimes X,\ R_{\mathbb{1}} : X \mapsto X \otimes \mathbb{1}$$

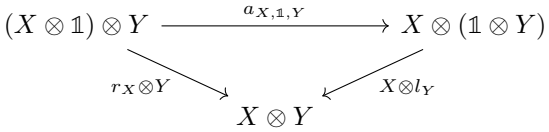
   are equivalences. We call this the *unit axiom*.

5.2 Remark.

- If we are only given a tensor product and associator $(\otimes, a)$ which satisfy the pentagon axiom, then in fact the existence of a pair $(\mathbb{1}, \iota)$ is a property and not any additional structure. More precisely, a unit pair is unique up to a unique isomorphism preserving its multiplication ι.
- Using the functors $L_{\mathbb{1}}$ and $R_{\mathbb{1}}$, we can get natural isomorphisms

$$r : - \otimes \mathbb{1} \to -, \text{ and } l : \mathbb{1} \otimes - \to -$$

for which the *triangle axiom* holds, which is the commutativity of the following diagram:



The following equivalence is proved in [EGNO15, Section 2.2].

5.3 Theorem. *Given a structure $(\mathcal{C}, \otimes, a)$ satisfying the pentagon axiom, then the data of $(\mathbb{1}, \iota)$ satisfying the unit axiom is equivalent to the data $(\mathbb{1}, l, r)$ satisfying the triangle axiom.*

We also use the following coherence theorem; see [EGNO15, Theorem 2.9.2].

5.4 Theorem (**Mac Lane's Coherence Theorem**). *In a monoidal category, given any two parenthesizations P_1 and P_2 of the object $X_1 \otimes \cdots \otimes X_n$, then any two isomorphisms $P_1 \to P_2$ obtained by using associators, their inverses, and tensor products with identity morphisms are actually the same.*

This extends the pentagon axiom to products of arbitrary length and allows us to suppress parentheses in tensor calculations.

Here are some examples of monoidal categories.

5.5 Example. Take Vect_k with the usual tensor product $\otimes_k = \otimes$. Then the associator is the one which "ignores parentheses":

$$a_{X,Y,Z} : (X \otimes Y) \otimes Z \to X \otimes (Y \otimes Z).$$

5.6 Example. Now, take A to be a Hopf algebra and consider the category $A - \mathrm{mod}_k$, where the tensor product is given by tensoring the underlying k-vector spaces, and A acts on $X \otimes_k Y$ by

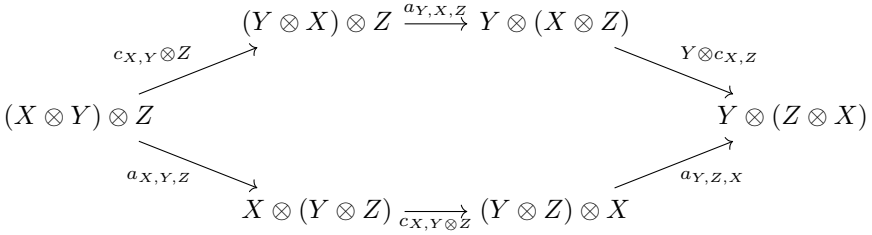
$$a(x \otimes y) = \sum a_1 x \otimes a_2 y,$$

where $\Delta(a) = \sum a_1 \otimes a_2$. The associator is inherited from Vect_k, and the unit is k with action $a \cdot t = \varepsilon(a)t$. Coassociativity gives associativity of the tensor action, while the counit identities give the unit constraints. In fact, a bialgebra suffices for this example.

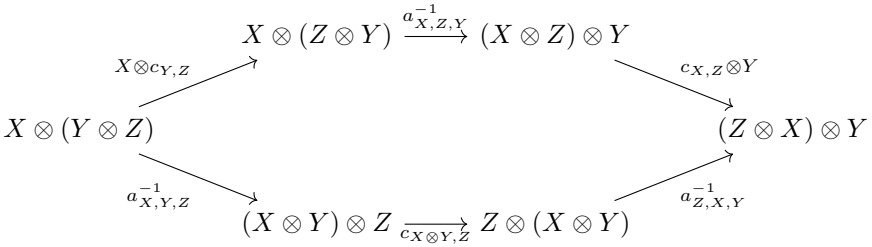
5.7 Definition. A *braided structure* on a monoidal category $(\mathcal{C}, \otimes, a)$ is a natural isomorphism

$$c_{X,Y} : X \otimes Y \to Y \otimes X$$

satisfying two hexagon identities, which are the following commutative diagrams:



and



Suppressing associators, the hexagon identities read

$$c_{X,Y\otimes Z} = (\mathrm{id}_Y \otimes c_{X,Z})(c_{X,Y} \otimes \mathrm{id}_Z),$$

$$c_{X\otimes Y,Z} = (c_{X,Z} \otimes \mathrm{id}_Y)(\mathrm{id}_X \otimes c_{Y,Z}).$$

The associators in the diagrams make these formulas meaningful without a strictness assumption.

5.8 Example. In Vect_k, the braid map is the obvious one:

$$c : X \otimes Y \to Y \otimes X,\ x \otimes y \mapsto y \otimes x.$$

5.9 Example. In $A-\mathrm{mod}_k$, the underlying braiding map of Vect_k lifts to one in $A-\mathrm{mod}_k$ exactly when the flip is A-linear; this holds when $\sum a_1 \otimes a_2 = \sum a_2 \otimes a_1$, namely when A is cocommutative. Conversely, if the flip is A-linear for all modules, apply it to $1\otimes 1$ in the tensor square of the regular module to obtain cocommutativity. For finite-dimensional modules this converse applies when A is finite dimensional. For a particular instance when we can do this, take $A = kG$, the group algebra for some group G. Then for any two G-modules $X, Y \in G-\mathrm{mod}$, then we can place a G-module structure on $X \otimes_k Y$ by

$$g(x \otimes y) = gx \otimes gy,$$

since the comultiplication on $g \in G$ is grouplike by construction. This also is a cocommutative Hopf algebra, and so the underlying braiding from Vect_k lifts.

More generally, a quasitriangular structure on a Hopf algebra gives a braiding on its module category. When A is finite dimensional, all braidings on its finite-dimensional module category arise this way; see [EGNO15, Section 8.3].

References

[BEO20] Dave Benson, Pavel Etingof, and Victor Ostrik. New incompressible symmetric tensor categories in positive characteristic, 2020.

[CEO23] Kevin Coulembier, Pavel Etingof, and Victor Ostrik. On Frobenius exact symmetric tensor categories. *Annals of Mathematics*, 197(3):1235–1279, 2023.

[CEO24] Kevin Coulembier, Pavel Etingof, and Victor Ostrik. Incompressible tensor categories. *Advances in Mathematics*, 457:109935, 2024.

[CF24] Kevin Coulembier and Johannes Flake. Towards higher Frobenius functors for symmetric tensor categories, 2024. The course discussion follows the May 2024 version.

[Del02] Pierre Deligne. Catégories tensorielles. *Moscow Mathematical Journal*, 2(2):227–248, 2002.

[DM82] Pierre Deligne and James S. Milne. Tannakian categories. In *Hodge Cycles, Motives, and Shimura Varieties*, volume 900 of *Lecture Notes in Mathematics*, pages 101–228. Springer, 1982. Corrected version, 2022, with the original proposition numbering.

[EGNO15] Pavel Etingof, Shlomo Gelaki, Dmitri Nikshych, and Victor Ostrik. *Tensor categories*, volume 205 of *Mathematical Surveys and Monographs*. American Mathematical Society, Providence, RI, 2015.

[ENO05] Pavel Etingof, Dmitri Nikshych, and Viktor Ostrik. On fusion categories. *Annals of Mathematics*, 162(2):581–642, 2005.

[EO19] Pavel Etingof and Victor Ostrik. On semisimplification of tensor categories, 2019.

[Mil12] James S. Milne. Basic theory of affine group schemes, 2012. Available at www.jmilne.org/math/.

[Ost15] Victor Ostrik. On symmetric fusion categories in positive characteristic, 2015.

[Wat79] William. C. Waterhouse. *Introduction to Affine Group Schemes*. Springer Graduate texts in mathematics. Springer New York, NY, 1979.

[Wib25] Michael Wibmer. A remark on torsors under affine group schemes. *Transformation Groups*, 30:447–454, 2025.